

High-speed silicon modulator based on cascaded microring resonators

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Abstract: A high-speed silicon modulator based on cascaded double microring resonators is demonstrated in this paper. The proposed modulator experimentally achieved 40 Gbit/s modulation with an extinction ratio of 3.9 dB. Enhancement of the modulator achieves with an ultra-high optical bandwidth of 0.41 nm, corresponding to 51 GHz, was accomplished by using cascaded double ring structure. The described modulator can provides an ultra-high-speed optical modulation with a further improvement in electrical bandwidth of the device.

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1. Introduction

Silicon photonics is a promising solution for future high-speed on-chip optical interconnects, enabling high-performance, cost-effective optical communication and computing systems [1–4]. Silicon electro-optical modulators have attracted a great deal of attentions over the years as they are the critical photonic components for generation and transmission of high-speed optical signals. To achieve high-speed optical signals modulation and transmission, modulators are required to operate with high optical bandwidth, high speed and high modulation efficiency. The microring modulator is one of several silicon electro-optical modulators that are designed to achieve these goals.

A great deal of efforts have been made to develop such high-speed silicon microring modulators. Through those efforts, dramatic improvements have been seen over the past decade with modulation speed of a silicon microring modulator improving from a few Gbit/s to over 30 Gbit/s [5–14]. The constant improvements are result of optimization in both the electrical and optical characteristics of the device. In the electrical improvement, special driving schemes [6, 7] and novel electrical structures, such as carrier-depletion effect in a reversed PN diode [8–11], are employed to improve the modulation speed. However, for the most popular refractive-index-modulation-based modulators, it is shown that one of the fundamental hurdles to the modulation speed is attributed to the optical features [15–17]. For an index modulated single microring, a sharp resonance (high Q factor) is required to achieve high dynamic extinction ratios with a relatively small resonance shift. However, a high-Q resonator has a longer photon lifetime which limits the achievable modulation bandwidth to the resonance linewidth when the resonance shift is comparable to the resonance linewidth. To relieve the Q-limited problem, coupled-ring-resonator and dual-ring structure are proposed to be used in enhancing the photon-lifetime-induced optical bandwidth for high-speed silicon modulator [15, 18]. However, these works on multi-ring modulators still remain at simulation stage. Though multi-ring structures have been investigated for wide bandwidth electro-optic switches [19, 20] and optical delay lines [21, 22], to the best of our knowledge, there has been no demonstration of high performance multi-ring modulators beyond theoretical stage.

In this work, we demonstrated what we believe to be the first experimental realization of a high-speed cascaded microring-based silicon modulator. The cascaded microring modulator consisted of two microrings structured in the SCISSOR configuration. The two microrings were optimized to provide much wider optical bandwidth and steeper sideband than a single microring and have large extinction ratio. Both of the two microrings were embedded with interleaved PN junctions so as to synchronously shift the broadened resonance and to form a depletion-mode modulator. The fabricated device provided a 3-dB bandwidth of 0.41 nm

(corresponding to 51 GHz) and high-speed modulation up to 40 Gbit/s with an extinction ratio of 3.9 dB. A high extinction ratio of 8.6 dB at 20 Gbit/s was also obtained.

2. Principle, design and fabrication

In the design of a microring-based modulator, a sharp resonance with a large extinction ratio (high Q factor) is usually employed so that a small resonance shift can be converted to a large amplitude modulation. Our simulation as illustrated in Fig. 1(a), indicated that a small resonance shift of 0.075 nm can provide extinction ratio of 19 dB. Such a small resonance shift only requires low driving voltages and, therefore, achieves low power consumption by the modulator. However, a high-Q resonator has a longer photon lifetime which limits the achievable modulation bandwidth to the resonance linewidth. As shown in Fig. 1(a), the 3-dB linewidth was only 0.15 nm, corresponding to 18 GHz. In above simulation, add-drop single ring was used for comparison with the later double channel cascaded double ring structure.

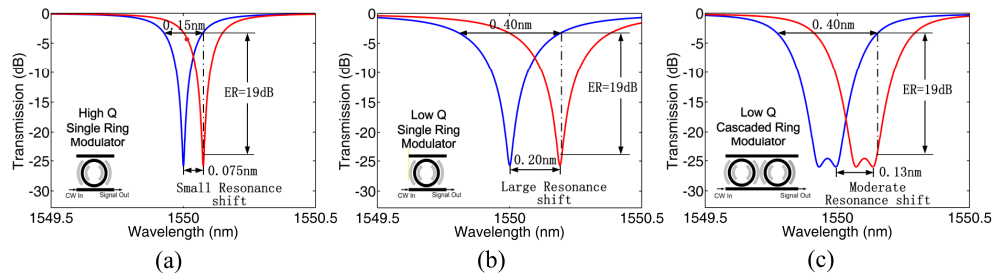


Fig. 1. (a) Single ring modulator scheme with a high Q factor microring. (b) Single ring modulator scheme with a low Q factor microring. (c) The proposed cascaded ring modulator scheme. In the simulation, the radius of the microring is 20 μm ; the effective refractive index of the ring waveguide is 2.91; the amplitude loss in the ring is $\alpha = 0.995$; the coupling coefficients at the through-port and drop-post are set to be $t_1 = 0.9615$, $t_2 = 0.97$ for (a), $t_1 = 0.903$, $t_2 = 0.9167$ for (b) and $t_1 = 0.9310$, $t_2 = 0.9555$.

One way to overcome the Q-limited problem is to use a low Q factor resonator with large extinction ratio. This type of resonance is attainable by increasing the coupling coefficient at the through-port and by making the coupling coefficient at the drop-port approaching the critical coupling. As shown in Fig. 1(b), the low Q resonator had a large 3-dB bandwidth of 0.40 nm. To achieve this, however, it required a large resonance shift to produce large extinction ratio since the large bandwidth made the side walls of the resonance less steep—this would require resonance shift of 0.2 nm to produce the extinction ratio of 19 dB.

Instead of using a single ring, we developed a double channel cascaded microring structure to afford large bandwidth and steep side walls of the resonance. As shown in Fig. 1(c), the bandwidth of the resonance was broadened to 0.40 nm, and a moderate resonance shift of 0.13 nm could provide extinction ratio of 19 dB. For the double channel cascaded microring structure, two factors contribute favorably to the desired spectra shape. One is the Bragg resonance between the two rings, which facilitates the broadening and deepening of resonance; the other is the introduced resonance spacing to distribute the resonances of the two rings, which helps to achieve wider bandwidth. In essence, we used a moderate coupling strengthen to achieve steep side walls of the resonance, and Bragg resonance and resonance spacing to broaden the bandwidth. In our simulation, a resonance spacing of the two rings was set to be 0.076 nm. In the simulation comparison, all the extinction ratios were calculated from the un-shifted resonance edge of 3 dB straight to the shifted resonance edge to ensure the same operation loss. So, for a fixed extinction ratio, the cascaded two ring structure had the combined advantages of an acceptable resonance shift in Fig. 1(a) and a wide bandwidth in Fig. 1(b). It is worth noted that the bottom of the resonance of the cascaded double ring is broadened to be a flat area. This feature would be an additional benefit of the cascaded double ring modulator when the modulation driving voltage is high enough or the resonance shift is large enough. In

Resonance in a microring:
A microring resonator produces a sharp dip (or peak) in its transmission spectrum at certain wavelengths. The “side walls” of this resonance are the slopes on either side of the resonance dip.

Steep side walls:
Steep slopes mean the transmission changes rapidly with wavelength. This is desirable for modulators because a small resonance shift (caused by an applied voltage) produces a large change in transmission — i.e., high modulation efficiency.

this case, the cascaded double ring device can withstand some shift without the laser tipping over to the other side of the resonance and spoiling the modulation depth.

For the improvement of electrical characteristics, we employed the carrier depletion-mode in our cascaded microring modulator design (the highest speed scheme for silicon-based modulators). In addition, the periodically interleaved PN junctions [23] were used to increase the overlap of depletion region and optical mode to achieve higher modulation efficiency. A doping concentration of $2 \times 10^{17} \text{ cm}^{-3}$ was chosen to obtain the highest modulation efficiency while keeping doping induced absorption loss low. The lengths of both p and n region are 300 nm, making a 600 nm long doping period. Interleaved width of the cascade PN junctions was specially optimized to 700 nm to ensure a low capacitance with a large misalignment tolerance as described in our previous work [12]. The partially enlarged view of the device in Fig. 2 shows the specially designed misalignment-tolerant interleaved PN junctions.

The cascaded microring modulator was fabricated on silicon-on-insulator (SOI) wafer with a 2- μm -thick buried oxide using a commercial 0.18 μm CMOS process from the Semiconductor Manufacturing International Corporation (SMIC) in China. Figure 2 shows a top-view of microscopic picture of the fabricated device and a 3D schematic view of the interleaved PN junctions. The bus and ring rib waveguides had a height of 340 nm and a slab thickness of 80 nm. Their widths were 450 nm and 500 nm, respectively. The radius of the two identical microrings was chosen to be 20 μm to provide sufficient length for the interleaved PN junctions to afford enough resonance shifts under a reverse voltage bias. The highly doped concentration was $1 \times 10^{19} \text{ cm}^{-3}$ for both p++ and n++ regions. For comparison, a single ring modulator with both the same electrical and optical parameters was also fabricated on the same wafer.

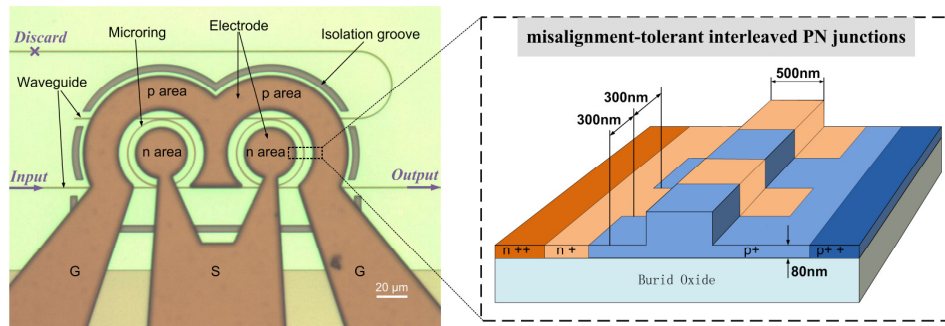


Fig. 2. Top-view of microscopic picture of the fabricated cascaded microring modulator and the schematic 3D view of the interleaved PN junctions.

3. Experimental Results and Discussions

Figure 3(a) and 3(b), respectively, show the measured normalized transmission spectra of the single ring modulator and cascaded double ring modulator with different DC voltages. The applied bias voltages varied from forward 0.5 V to reverse 8 V. The Q factor of the single ring resonator was estimated to be approximately 5600 for zero bias. The 3-dB bandwidth of the single ring was 0.28 nm and broadened to 0.41 nm by the double ring structure. The static extinction ratio is enhanced from 16 dB to 26 dB. As can be seen in the two figures, the cascaded double ring structure had a much steeper sideband than that of the single ring, with the right sideband especially steeper since there is a slight asymmetry of the spectra. This asymmetry, we believe, stems from deviation from the original design as the result of the fabrication process. Red shift of the resonance was obtained due to carrier depletion with the increase of reverse bias. The single ring modulator, from 0.5 V forward bias to -6 V reverse bias, displayed a resonance shift of 90 pm which affords 11.7 dB modulation depth and 5.7 dB insertion loss. The cascaded double ring modulator, on the other hand, showed a resonance shift of 74 pm, resulting in 19.7 dB modulation depth and 4.4 dB insertion loss. Enhanced performance is, therefore, achieved with the modulator based on cascaded microring structure.

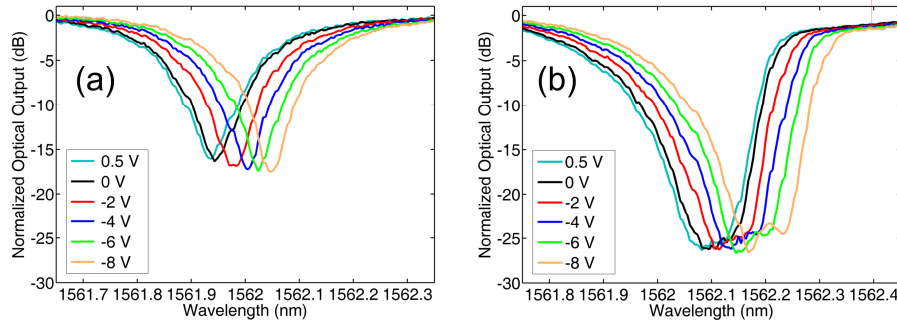


Fig. 3. Normalized transmission of the fabricated single ring modulator (a) and cascaded double ring modulator (b) for different applied reverse bias from 0.5 V to -8 V.

The effective index variations and the figure of merit $V_{\pi}L_{\pi}$ of the active region were calculated from the wavelength resonance shifts and are reported in Fig. 4. It shows that the effective index change of the double ring modulator is little smaller than that of the single ring. Theoretically, however, the effective index change should be the same as the resonance shifts are determined by ring's dimension and doping parameters, both of which were kept the same for single and double ring modulator. We believe that the fabrication-induced and measurement-induced variations are the root cause for such abnormality. Correspondingly, the $V_{\pi}L_{\pi}$ showed a change from 1.5 to 2.2 V·cm and from 2.1 to 2.6 V·cm for single ring and double ring modulator, respectively.

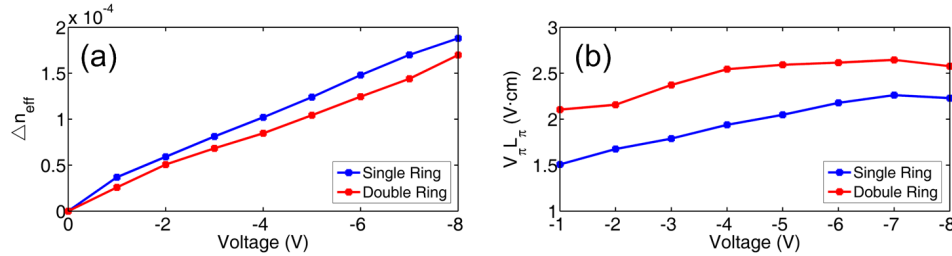


Fig. 4. (a) Effective index variation as a function of the reverse bias. (b) $V_{\pi}L_{\pi}$ as a function of the reverse bias. Both single ring and cascaded double ring modulator's performances are included.

The high-speed characteristics of the ring modulator can be investigated using a circuit model extracted by curve-fitting the measured S11 data [13,14]. The S11 magnitude and phase response was measured from DC to 30 GHz by a signal integrity network analyzer (SPARQ, Lecroy). Figure 5 shows the measured S11 data and the curve fitting results from the equivalent circuit model. In the circuit model, C_p represented the capacitance between the electrodes through the top dielectrics and the air, and C_j and C_{ox} were the capacitances in the reverse-biased diode junction and through the buried oxide layer, respectively. R_s and R_{si} are the resistance of the reverse-biased PN junction and the resistance of the substrate silicon layer, respectively. With the extracted circuit values at 0 V, the electrical bandwidth of the device under test was calculated to be 21 GHz when it was loaded with a 50 Ω source [14].

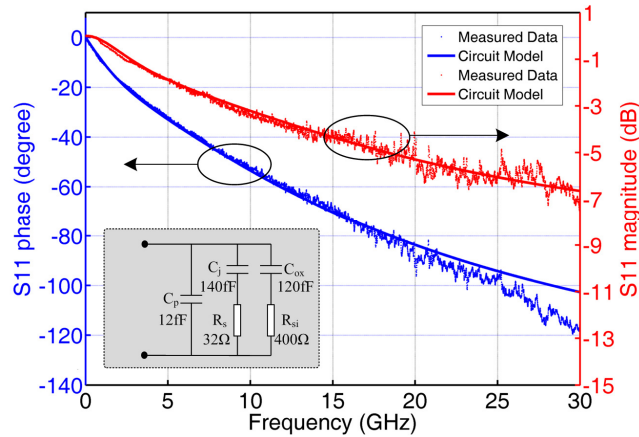


Fig. 5. Measured S11 magnitude (red) and phase (blue) responses with corresponding fitting curves simulated from the equivalent circuit model.

High-speed performance of the fabricated cascaded double ring modulator was investigated by measuring optical eye diagrams at high bit rates. A Centellax pattern generator was used to obtain different bit rates of pseudorandom binary sequence (PRBS) signal with a $2^{15}-1$ pattern length. After a driver amplifier, the PRBS output signal was connected to a bias tee to add a DC bias to the RF signal, keeping a constant reverse-bias to the modulator diode. The DC and RF signals then applied to the ring modulator, using 50 GHz electrical ground-signal-ground probes. The output light from the modulator was amplified by Erbium-doped fiber amplifier (EDFA) and transmitted through a band pass filter. The optical signal was measured by a 65 GHz optical head on a Tektronix digital scope DSA8300. The obtained optical eye diagrams at bit rate of 20, 30 and 40 Gbit/s are shown in Fig. 6. The electrical voltage peak-to-peak of 5 V and DC bias of -4.5 V were used in our measurement. As demonstrated, wide open eyes at 20 and 30 Gbit/s were achieved. A high extinction ratio of 8.6 dB at 20 Gbit/s was obtained. As the bit rate increased to 30 and 40 Gbit/s, the extinction ratio dropped to 5.8 dB and 3.9 dB, respectively. Since the optical bandwidth of the cascaded microring structure is large enough (51 GHz), the deteriorating performance of the modulation, including the extinction ratio decreasing, is intrinsically limited by the electrical bandwidth of 21 GHz as calculated above. In addition, there is a non-ignorable reflection of the electrical driving signal while the device is not an exact 50 Ω matched termination. The reflection would become large while it was used a relatively large driving voltage during the measurement. So the large reflection would distort the clock signal so that it increases the time jitter of the PRBS source. Such time jitter would spoil the eye diagram especially at high speed. Therefore, it is expected that further improvement of electrical design of the cascaded ring modulator would be beneficial for further enhancement in the speed of modulation. In fact, the active area of the cascaded double ring modulator was two times of that of the single ring modulator, resulting in a larger capacitance and a smaller resistance. Therefore, further work to optimize cascaded ring modulator to obtain the optimal electrical bandwidth will be needed.

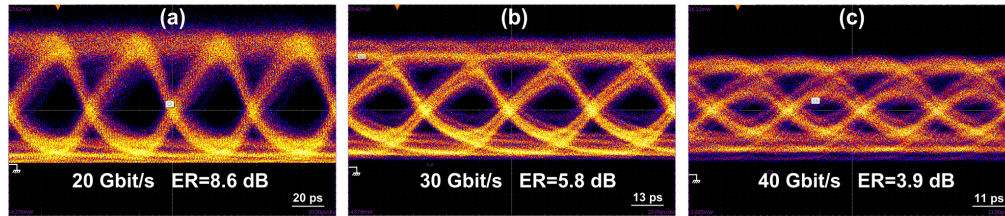


Fig. 6. Measured (a) 20 Gbit/s, (b) 30 Gbit/s and (c) 40 Gbit/s eye diagrams of the cascaded ring modulator under the driving voltage of 5 V.

4. Conclusion

In conclusion, we have fabricated and demonstrated a high-speed cascaded microring-based silicon modulator based on interleaved PN junctions. The cascaded double ring structure shows advantages of a large optical bandwidth and a steeper sideband over a single ring with large extinction ratio. The measured 3-dB bandwidth was 0.41 nm, which corresponded to 51 GHz. High-speed modulation up to 40 Gbit/s with extinction ratio of 3.9 dB was also experimentally achieved at drive voltage of 5 V. A high extinction ratio of 8.6 dB was reached at 20 Gbit/s. The large optical bandwidth of the cascaded double ring modulator has set the ball of further increasing the modulation speed squarely in the court of electrical improvement. It is expected that a much higher speed of modulation can be achieved with such cascaded microring modulator by future improvement in the electrical characteristic of the component.

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